2D Graph Editor Program – Project Prompts

Prompt 1: Graph Creation

Develop a 2D Graph Editor that allows users to create graphs interactively by adding vertices and connecting them with edges. The application should support both directed and undirected graphs. Users should be able to place vertices anywhere on the canvas and create connections between them using simple mouse operations.

Prompt 2: Graph Editing

Design a graph editing system where users can modify existing graphs by moving vertices, deleting vertices or edges, and updating labels or weights. The editor should automatically update all related connections when a vertex is modified or removed, ensuring graph consistency.

Prompt 3: Graph Visualization

Create a graphical interface that visually represents graphs in a two-dimensional space. Vertices should be displayed as circles with labels, while edges should be represented as lines or arrows. Different colors and styles can be used to distinguish selected elements, weighted edges, and traversal paths.

Prompt 4: Graph Algorithms

Implement graph algorithms within the editor to help users analyze graph structures. Features should include Breadth-First Search (BFS), Depth-First Search (DFS), shortest path calculation, cycle detection, and connectivity analysis. Results should be displayed visually on the graph.

Prompt 5: Automatic Layout Generation

Develop an automatic layout feature that arranges graph vertices neatly on the canvas. The system should reduce edge crossings and improve readability by positioning vertices intelligently. Users should also have the option to manually adjust the generated layout.

Prompt 6: File Management

Create functionality for saving and loading graph projects. Users should be able to store graph data in files and reopen them later for further editing. The editor should also support exporting graphs as images for presentations and reports.

Prompt 7: Educational Tool

Build the 2D Graph Editor as an educational platform for learning graph theory and data structures. The program should visually demonstrate graph traversal algorithms and shortest path computations, helping students understand graph concepts through interactive simulations.

Prompt 8: Advanced Graph Analysis

Develop advanced analytical features that calculate graph properties such as degree distribution, connected components, graph density, and minimum spanning trees. The editor should present these statistics in an easy-to-understand format alongside the graphical representation.

Prompt 9: User-Friendly Interface

Design a simple and intuitive user interface that enables users to create and edit graphs without technical expertise. The application should provide toolbars, context menus, keyboard shortcuts, and real-time feedback for all graph operations.

Prompt 10: Complete Project Description

The 2D Graph Editor is a software application that enables users to create, edit, visualize, and analyze graph structures in a two-dimensional environment. It provides interactive tools for vertex and edge management, supports graph algorithms, offers automatic layout generation, and allows project saving and exporting. The system serves as both a practical graph-design tool and an educational platform for understanding graph theory concepts.